23 Bayesian Simple Linear Regression Model

Regression is one of the most widely used statistical techniques for modeling relationships between variables. Regression is also widely used in prediction. We will now consider a Bayesian treatment of *simple linear regression*. We'll use the following example throughout.

Percent body fat (PBF, total mass of fat divided by total body mass) is an indicator of physical fitness level, but it is difficult to measure accurately. Body mass index (BMI), which is easily computed based on height and weight (BMI $= \frac{\text{weight in kg}}{(\text{height in m})^2}$), is another commonly used measure[1]. What is the relationship between BMI and percent body fat? How can we use a person's BMI to estimate their percent body fat?

We'll fit a Bayesian model for predicting percent body fat from BMI using data on a nationally representative sample of 2157 U.S. males from the National Health and Nutrition Examination Survey (NHANES).

Example 23.1. Consider the assumptions of a Bayesian simple linear regression model for predicting percent body fat from BMI.

1. What are the assumptions of the simple linear regression model? (There are many different regression models. We're referring to the one that is usually the first one you see in introductory statistics. Sometimes the assumptions are summarized with the acronym "LINE".)

2. What are the parameters in the model? That is, what do we need to put a prior distribution on (in order to eventually get a posterior distribution given sample data)? What do the parameters represent in context?

[1]But BMI has its own problems; see http://fivethirtyeight.com/features/bmi-is-a-terrible-measure-of-health/.

3. Suggest a prior distribution. Explain your reasoning.

4. Describe in full detail how you would simulate the prior predictive distribution of percent body fat for males with a BMI of 20.

5. Simulate the prior predictive distribution of percent body fat for a range of BMI values. Do the distributions seem reasonable? If not, experiment with different priors until you find one that produces a reasonable prior predictive distribution.

- Consider data on a numerical response variable y and a numerical explanatory (or predictor) variable x. A **simple linear regression model** of y on x assumes[2]:
  - *Linearity*: the conditional mean of y is *linear* in x; for some slope β_1 and intercept β_0:
  $$\mathrm{E}(y|x) = \beta_0 + \beta_1 x$$
  - *Independence*: $y_1, \ldots, y_n$ are independent[3]
  - *Normality*: for each x, the conditional distribution of y given x is Normal
  - *Equal variance*: the conditional variance of y does not depend on x
  $$\mathrm{Var}(y|x) = \sigma^2$$
- The regression model is *conditional* on x. That is, we make assumptions about and perform inference for the conditional distribution of the response variable y given values of the explanatory variable x. Therefore, even when we technically have a sample of pairs (x_i, y_i) from a bivariate distribution, we will always condition on the values of x and treat them as non-random.

[2]In a Bayesian context, all of these assumptions are also *conditional* on the parameters β_0, β_1, σ

[3]In a Bayesian context, we assume conditional independence of $y_1, \ldots, y_n$ given the parameters β_0, β_1, σ

- Linearity is an assumption: we assume there is some "true" population *regression line* $\beta_0 + \beta_1 x$ which specifies the conditional *mean* of the response variable for each value of the explanatory variable.
- The above set of assumptions can be stated more compactly as

$$y_i \sim N(\beta_0 + \beta_1 x_i, \sigma), \text{ and } y_1, \dots, y_n \text{ are independent}$$

  Equivalently,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \text{ where } \varepsilon_i, \dots, \varepsilon_n \text{ are i.i.d. } N(0, \sigma).$$

- The parameters of the above model are β_0, β_1, and σ.
- σ_y^2 denotes the variance of the response variable y with respect to its overall mean, while σ^2 denotes the variance of values of the response variable about the true regression line. Thus, $\sigma^2 = (1-\rho^2)\sigma_y^2$, where ρ is the population correlation between the explanatory and response variables.

Example 23.2. Section 23.1.3 summarizes the sample data. Before fitting a Bayesian model, we'll examine the usual frequentist results.

1. Find and interpret the slope of the fitted regression line.

2. Find and interpret a 95% confidence interval for the slope of the regression line. (How do you interpret 95% confidence?)

3. Find the equation of the fitted regression line.

4. Find and interpret the fitted value for men with a BMI of 20 kg/m^2.

5. Find and interpret a 95% confidence interval based on the fitted value for men with a BMI of 20 kg/m^2.

6. Find and interpret a point estimate of σ.

7. Find and interpret a 95% *prediction* interval for men with a BMI of 20 kg/m^2.

Example 23.3. Now we'll fit a Bayesian model and find the posterior distribution.

1. Suppose there are just two observations of (BMI, PBF): (15, 20) and (40, 30). How would you compute the likelihood?

2. Use `brms` to fit the model to the full sample data. Run some diagnostic checks; does the MCMC algorithm seem to be working efficiently?

3. Inspect the joint posterior distribution of β_0 and β_1. What do you notice? Why do you think this happens?

4. Now suppose we replace BMI (x) with its centered version ($x - \bar{x}$). What does the parameter β_0 represent in this model? Suggest a prior distribution for β_0 (considering it in isolation).

5. Use `brms` to fit the model using centered BMI as the explanatory variable. Inspect the joint posterior distribution of β_0 and β_1. What do you notice? Why do you think this happens?

- It is often recommended to center predictors in regression models.
- Centering can help interpretation (because the intercept is often meaningless in context), and also help with multicollinearity (in polynomial models, or when there are multiple predictors).
- In Bayesian framework, centering can reduce the correlation between parameters amd improve efficiency of MCMC methods
- When there are multiple numerical predictors it is also often recommended to standardize them (by subtracting the mean and dividing by one or two standard deviations). In a Bayesian framework, standardizing can help set priors when the predictors are on different scales.

Example 23.4. Now we'll summarize and report conclusions based on the posterior distribution, using the model with centered BMI.

1. Describe the posterior distribution of β_1. Find and interpret 50%, 80%, and 98% posterior credible intervals for β_1.

2. What does the parameter $\mu_{20} = \beta_0 + \beta_1(20 - \bar{x})$ represent in context? Explain how you could approximate the posterior distribution of μ_{20}. Then approximate the posterior distribution, describe it, and find and interpret a 98% posterior credible interval for μ_{20}.

3. Describe the posterior distribution of σ. Find and interpret 50%, 80%, and 98% posterior credible intervals for σ.

4. Compare the results of the Bayesian and frequentist analyses. How are they similar? How are they different?

Example 23.5. Now we'll use the fitted model to predict percent body fat.

1. Explain how you could approximate the posterior predictive distribution of percent body fat for men with a BMI of 20 kg/m^2.

2. Find and interpret a 95% posterior prediction interval for men with a BMI of 20 kg/m^2.

3. Compare the 95% posterior prediction interval with the corresponding frequentist 95% prediction interval. How are they similar? How are they different?

4. Suggest how you could perform a posterior predictive check. Then use software to perform the check. Does the model seem reasonable based on the data?

- There are a number of ways the assumptions of the simple linear regression model could be revised.
- We could assume the mean of y changes with x is a non-linear way (e.g., quadratic).
- We could assume distributions other than Normal in the likelihood. For example, we could assume a t-distribution if outliers were an issue.
- We could assume the conditional variance of y also changes with x.
- We could include multiple explanatory variables x.
- Each of the above would be a change in the *likelihood*, perhaps introducing additional parameters.

Example 23.6. Now we'll investigate sensitivity of the results to choice of prior. Rerun the model (using centered BMI) but let `brms` choose the prior. Compare the results of the two Bayesian models; are the results very sensitive to the choice of prior?

23.1 Notes

23.1.1 Prior predictive tuning

- We'll start with a prior distribution under which β_0, β_1, and σ are independent.